Name: ______________________________

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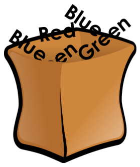
Modeling Language with AI

Another kind of AI, called **Generative AI**, is all about making new text, images, or music. The tools that generate language are called ***language models***, and they power everything from predictive text to chatbots, and from deepfakes to AI-written book reports. But a computer can't actuallty *read* : it has no idea what words **mean**. So how can it possibly work with language?
The trick, as always, is to turn language into **data**. Once words become data, a computer can do two big jobs with them:

Understanding Text	Generating Text
 A ***bag of words*** represents a document as an **unordered** collection of words and how often each appears. Order is thrown away — just the counts remain. Now text is data, so we can measure **similarity** between documents (this is how plagiarism detectors work!).	 A ***statistical language model*** learns, from a corpus, **how often one word follows another.** It can then predict — and generate — a likely next word, one word at a time.

Raw text is noisy, so before we model it we ***normalize*** it — lowercasing everything, stripping punctuation, and removing common ***stopwords*** (like "the" and "of"). This cuts out noise so the words that actually carry meaning produce more meaningful results.

Probability and N-grams

A statistical language model is built on ***probability***: the *fraction* of times one word follows another in the training corpus. To measure that, we chop the corpus into chunks called ***n-grams*** (where n is the number of words per chunk):

Name	Chunks of "There was an old lady"	What it captures
Unigram (n=1)	(There) (was) (an) (old) (lady)	just word frequencies — *a bag of words!*
Bigram (n=2)	(There was) (was an) (an old) (old lady)	which word tends to follow one other word
Trigram (n=3)	(There was an) (was an old) (an old lady)	the next word, given the *two* before it

Counting up these n-grams to compute next-word probabilities **is** training a model — and like all training, it's **lossy**: the model keeps the probabilities but forgets the original text.

Generating Text = Sampling

To generate, the model just **samples** a next word according to those probabilities, adds it to the sentence, and repeats. A dial called ***temperature*** controls how random the choice is: low temperature always grabs the most likely word (repetitive), while higher temperature takes more chances (surprising, sometimes nonsense). A tool like **Soekia** lets us peek behind the curtain at exactly this process.

A language model knows **probability**, not **meaning** or **truth**.
This is why language models "make stuff up," or **hallucinate**: every sentence is produced by the same word-by-word probability game, whether or not it happens to be true. For the same reason, a model faithfully echoes the **biases** baked into its corpus.

A "Language" is more than Just Words!

Natural language processing (NLP) means using a computer to consume a corpus, build a model, and produce new text. But nothing about that process requires *English* — or any human language at all! The very same techniques work on **artificial languages**: the moves of a Tic-Tac-Toe game, a melody, or a chess match are all just sequences of "words" a model can learn to continue.

Text Similarity

Save a copy of [AI Text Starter File] and click "Run". You may also want to review these Contracts:

```
# distance-similarity :: ( Table , String , List<String>) -> Table
                              t       ID       dimensions

# angle-similarity :: ( Table , String , List<String>) -> Table
                           t       ID       dimensions
```

What's Going on in this Code?

1) What is being defined in the first 22 lines of this file?

2) What happens on lines 25-38?

3) What is defined on line 40?

4) In the Interactions Area, evaluate `whale-essay`, then `whale-essay["EMOJI"]` and `essay-img(whale-essay)`. Explain what `essay-img` does.

5) Find line 22 and scan the text of the `mystery` document (`"❓"`). Which essay do you expect it to be *most similar* to? Why?

6) Why *can't* we use our `distance-similarity` and `angle-similarity` functions with the `corpus` table, as-is?

Describing Essays with Numbers

Pyret has a number of functions that compute numbers, describing different features of text:

```
# string-length :: String -> Number
# num-words :: String -> Number
# num-syllables :: String -> Number
```

```
# num-sentences :: String -> Number
# text-grade :: String -> Number
```

7) Evaluate `string-length(badger)`, and experiment with the other functions.

8) In the Definitions Area, define a new table: `decorated = decorate-text-table(corpus, "DOC")`. Click "Run" and evaluate `decorated`. What did you get back?

9) Evaluate `image-scatter-plot(decorated, "WORDS", "SYLLABLES", essay-img)` to see these essays in "WORDS-SYLLABLES Space". Then make another plot in "WORDS-SENTENCES Space". What do these scatter plots tell us?

10) Try using `distance-similarity` and `angle-similarity` with any combination of your new columns. Can you create a space where we can see that `"❓"` is closest to 🐘? Why or why not?

Normalizing Text

This page is designed for use with AI Text Starter File.

Counting Badgers

> *The American badger is a North American badger similar in appearance to the European badger, although not closely related. It is found in the western, central, and northeastern United States, northern Mexico, and south-central Canada to certain areas of southwestern British Columbia. The American badger's habitat is typified by open grasslands with available prey (such as mice, squirrels, and groundhogs).*

1) How many times does the word "badger" appear in this essay? ______________________________

2) Evaluate `computed` in the Interactions Area and find the `badger` column. How many did Pyret find? ______________

3) Why didn't Pyret find the same number of `badger` mentions that you did? ______________________________

____

4) What could Pyret do differently, to find all the badger references that you did? *Try to generate as many suggestions as you can. Look for other examples in the table where a word was improperly counted as separate.*

____

____

5) When humans scan text for similarity, there are many words that we tune out because they're used all the time. We call these ***stop words***. Find 10 ***stop words*** in the badger essay, to fill in the blanks below:

_______ _______ _______ _______ _______ _______ _______ _______ _______ _______

Normalizing Text in Pyret

Evaluate `vacation` in the Interactions Area.

> *"Vacation is fun! One of my favorite things about vacation is that I have time for breakfast. What do you love about vacation?*

6) Evaluate `lowercase(vacation)`. How many changes does `lowercase` make to the original string? ______________

Describe them: ____

7) Evaluate `remove-punct(vacation)`. How many changes does `remove-punct` make to the original string? ______________

Describe them: ____

8) Evaluate `remove-stop-words(vacation)`. Mark all of the changes on the string at the top. How many changes does

`remove-stop-words` make to the original string? _______ Describe what happened: ______________________________

____

★ What code will take `vacation` and remove punctuation *and* convert to lowercase? ______________________________

Why Normalizing Matters

9) Evaluate `all-cols-similarity(computed, "?")`. **Before normalizing the data…**

- which essay is most similar to `?` _______. What's the angle-similarity between them? ______________________
- what's the range of angle-similarity measures between `?` and the other documents? ______________________

10) Evaluate `all-cols-similarity(norm-computed, "?")`. **After normalizing the data…**

- which essay is most similar to `?` _______. What's the angle-similarity between them? ______________________
- what's the range of angle-similarity measures between `?` and the other documents? ______________________

Detecting Plagiarism using Similarity

Making Sense of the Space

A teacher asks students to write an essay on one of three topics: modern art, soccer, or zebras. The teacher feeds the essays to a plagiarism detector **to compare the essays to one another.** It plots 32 points in a 1023-dimensional space.

1) Why does it plot 32 points? ______________________________

2) Why does it generate a 1023-dimensional space? ______________________________

____

3) The teacher knows, of course, that students don't merely copy one another's writing; instead, they copy/paste text from the internet! The teacher opens her plagiarism detection software and tells it to consume a large ***training corpus*** of internet text about modern art, soccer, and zebras. How will the number of dimensions of the new space compare?

Circle one: more dimensions the same number of dimensions fewer dimensions

4) Explain your choice.

____

____

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Interpreting a Plagiarism Detectors Output

Match the description of each output with the correct interpretation.

Description of the Output			Interpretation of What the Output Means
The 32 points are evenly scatter all over the 1023-dimensional space, with no apparent clustering.	1	A	Essays about one topic are likely to use similar words, but essays about different topics likely use different words. The three clusters probably represent the three different topics (modern art, zebras, and soccer).
The points are clustered into three groups, but two of the points are almost on top of one another.	2	B	Two of the essays use almost the same words, in almost the same frequencies. This look suspicious, and the teacher should take a closer look at those essays for potential plagiarism.
The points are clustered into three groups.	3	C	Only one student opted to write about a particular topic, while the rest of the class wrote essays on the other two. There is no evidence of plagiarism.
Almost all the points are clustered into two groups, but one point is positioned very far away from both clusters.	4	D	Every student appears to have written essays about entirely different topics! The teacher has no reason to suspect plagiarism, but they should *probably* worry about their students not following the assignment!

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